

Lakshmipathiraju Pericharla

Data Scientist / ML Engineer

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PROFESSIONAL SUMMARY

- 5+ years of experience as a Machine Learning Engineer specializing in managing the entire ML lifecycle, from data collection to deployment and monitoring, ensuring robust and reliable ML services at scale.
- Proven expertise in collaborating with cross-functional teams, including Data Science, QA, and Infrastructure, to effectively productionize machine learning models and enhance operational efficiency.
- Strong proficiency in Python, SQL, and Java, with a solid understanding of algorithms and data structures, enabling the development of efficient and scalable machine learning solutions.
- Extensive experience with machine learning frameworks such as Pytorch, Tensorflow, and Sklearn, applying advanced techniques like regression, classification, and deep learning to solve complex business problems.
- Demonstrated ability to stay updated with the latest developments in data science and machine learning, integrating new technologies to evolve existing solutions and drive innovation.
- Skilled in big data technologies, including Spark and Airflow, with a comprehensive understanding of system architecture and data pipelines to support large-scale data processing.
- Excellent written, verbal, and presentation skills, capable of conveying complex concepts in a clear and simple manner to diverse audiences, enhancing stakeholder engagement.

SKILLS

Programming Languages: Python, Java, Scala, SQL, R, C++, JavaScript, TypeScript

Machine Learning Frameworks: TensorFlow, PyTorch, Scikit-learn, Keras, MLFlow, XGBoost, LightGBM, CatBoost

Big Data Technologies: Spark, EMR, Kafka, Airflow, Hadoop, Hive, Redshift, MongoDB

Machine Learning Models: LLMs, GNN, Deep Learning, Logistic Regression, Gradient Boosting Trees, Neural Networks, Sequence Models, Decision Trees

Cloud Platforms: AWS, S3, SageMaker, EC2, Lambda, Google Cloud Platform, Azure, Databricks

DevOps Tools: Docker, Kubernetes, Jenkins, Git, CI/CD, Terraform, Ansible, Grafana

Data Processing: Data Pipelines, ETL, Data Wrangling, Data Collection, Data Cleaning, Feature Engineering, Data Visualization, Streaming Architecture

Soft Skills: Communication, Collaboration, Problem-Solving, Adaptability, Critical Thinking, Time Management, Presentation Skills, Teamwork

System Architecture: Microservices, Monolithic Architecture, Event-Driven Architecture, API Design, Service-Oriented Architecture, Distributed Systems, Scalability, Reliability

Project Management: Agile Methodologies, Scrum, Kanban, Stakeholder Management, Requirements Gathering, Risk Management, Time Estimation, Resource Allocation

WORK EXPERIENCE

Scale AI - San Francisco, CA

Senior Data Scientist - Jun 2023 to Present

- Spearheaded the management of the entire Machine Learning (ML) lifecycle, from data collection to deployment and monitoring, utilizing Python and SQL to enhance model performance and reliability.
- Collaborated with cross-functional teams, including Data Science (DS), Quality Assurance (QA), and Infrastructure, to productionize ML models, ensuring seamless integration and operational efficiency.
- Optimized code for production environments, leveraging technologies like Docker and Kubernetes to ensure robustness and scalability of ML services across various applications.
- Developed and implemented advanced ML algorithms, including Gradient Boosting Trees and Neural Networks, to solve complex business problems, resulting in a 30% increase in model accuracy.
- Managed and evolved existing solutions by incorporating cutting-edge technologies such as Pytorch and TensorFlow, enhancing the capabilities of ML applications in real-time data processing.

- Conducted thorough presentations and documentation to convey complex ML concepts clearly to stakeholders, fostering better understanding and collaboration across teams.
- Stayed abreast of the latest developments in data science and machine learning, integrating new methodologies to drive innovation and maintain competitive advantage.
- Enhanced data pipelines and streaming architecture using technologies like Apache Kafka and Spark, improving data processing efficiency by 40%.
- Implemented monitoring solutions with Grafana and Datadog to track model performance and system health, ensuring high availability and reliability of ML services.
- Mentored junior data scientists, providing guidance on ML best practices and fostering a culture of continuous learning and improvement within the team.

Technologies Used: Python, SQL, Pytorch, TensorFlow, Docker, Kubernetes, Apache Kafka, Spark, Grafana, Datadog, MLFlow, AWS, Redis, MongoDB, Airflow, S3, Jenkins

American Express - New York, NY

Machine Learning Engineer II - Jun 2020 to Dec 2022

- Engineered and deployed machine learning models for fraud detection, utilizing Logistic Regression and Deep Learning techniques to enhance transaction security and reduce false positives by 25%.
- Collaborated with data engineering teams to design and optimize data pipelines, leveraging Apache Airflow and EMR to streamline data processing and improve model training times.
- Developed RESTful APIs using Flask and FastAPI to facilitate seamless integration of ML models into existing applications, enhancing user experience and operational efficiency.
- Conducted extensive data analysis and feature engineering, applying statistical techniques to derive insights and improve model performance across various business units.
- Managed the deployment of ML models using AWS Sagemaker, ensuring scalability and reliability in a cloud environment, which supported over 1 million transactions daily.
- Implemented monitoring and logging solutions with Kibana to track model performance metrics and troubleshoot issues proactively, maintaining high service levels.
- Participated in Agile methodologies, contributing to sprint planning and daily standups to ensure timely delivery of projects and alignment with business goals.
- Researched and integrated large language models (LLMs) and prompt engineering techniques to enhance personalization features, resulting in a 20% increase in customer engagement.
- Trained and mentored new team members on ML lifecycle best practices, fostering a collaborative environment and enhancing team capabilities.
- Presented findings and model outcomes to senior management, effectively communicating complex technical concepts in an accessible manner.

Technologies Used: Python, SQL, AWS, Flask, FastAPI, Apache Airflow, EMR, Sagemaker, Kibana, Docker, Redis, Spark, Kafka, TensorFlow, Pytorch, MLFlow, Jenkins

CERTIFICATIONS

- AWS Certified Machine Learning - Specialty
- Google Professional Machine Learning Engineer
- Databricks Certified Machine Learning Professional
- Certified Analytics Professional (CAP)

EDUCATION

Master's in Information Technology | Clark University, Worcester, MA | 3.6 GPA